ANALYSIS OF FINANCIAL TIME SERIES USING DEEP LEARNING



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DECLARATION

I hereby declare that the work presented in this thesis is my individual work, and has been done under the supervision of Kavindi Liyanage. Every work presented in this paper has been cited appropriately. I declare that this thesis has not been submitted for the award of any degree or diploma at any other university.

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This thesis can therefore be attributed to all the efforts that I, as well as all the members involved in this process, have made. Thank you for, motivating me and encouraging me in all my endeavours.

ABSTRACT

An appropriate determination, estimation, and forecasting of variables in the field of finance and specifically when it is regarding stock prices, currency exchange rates, and markets require the understanding of the properties and characteristics of financial time series. This paper provides a comparison of deep learning based paradigms namely LSTM and CNN with conventional methods like ARIMA and GARCH.

These methods are concerned with improvement in the predictive accuracy, stability and speed of converging to a solution through applying benchmark datasets from the financial domain for research amplification. Data pre-processing tools like normalization and feature transformation were used in an effort to get the data in the best shape as possible prior to entering it into the model. Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) were used as performance measures in order to analyze the performance of the compared models.

Findings show that deep learning models are more effective than previous paradigms for modelling non-linear and long-term temporal structures typical for financial time series data. It has been found that LSTM models are commonly used in the sequential representation and CNNs are good at feature extraction. Nonetheless, simple methods still perform comparably well in low-data and low-computing power environments.

The comparative analysis offers insights of comparing one model against another highlighting the fact that before conducting the research one needs to choose the most appropriate methodology depending on the particular cases and limitations. The present work makes a contribution to the existing research on the application of deep learning techniques in the context of finance and the discussed ideas indicate the directions for their further integration with classical models in order to boost the accuracy of financial forecasts.

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Chapter 1: Introduction

1.1 Background of the Research

As markets evolve and financial decisions increase in complexity, accurate forecasts of financial time-series provide significant value both to financial institutions and investors, as well as policymakers. Heterarchical heuristics having a simplistic characteristic, including ARIMA, GARCH, and Exponential Smoothing, have been implemented in the market trend forecast for several decades (Sutcu and Gulbahar, 2023). Although valuable in some respects, these models fail to address the complexity, nonlinearity, and size of data characteristic of today's financial markets. Besides, they are not capable of modeling long-term dependencies and other complex structures inherent in data for predicting values in shift or high variability market environments.

ML BNs have recently been applied in managing large datasets, identifying non-linear patterns, and learning temporal dependencies for models (Miller *et al.* 2024). Among RNNs, the LSTM networks have been proved to be the most suited to face the difficulties involved in the processing of sequential data such as time series data on account of their long memory capability. CNN, which used to be employed in image processing, also found application in time series forecasting when data is converted into the appropriate format and can help the network to sketch complex patterns.

1.2 Research Aim

The main research purpose is to investigate the efficiency of deep learning models ininese. In particular, LSTM and CNN in the prediction of financial time series data. The goal is to create a highly reliable and accurate predictive model that would improve precision of stock prices' forecasts and help traders and investors with their decision-making.

1.3 Research Objectives & Questions

Objectives

- To develop as well as evaluate the performance of the LSTM & the CNN models while improving the accuracy of financial time-series forecasting.
- Compare the effectiveness of LSTM & the CNN models with the traditional forecasting techniques, especially ARIMA, GARCH, as well as Exponential Smoothing, while capturing the complex patterns in case of financial data.

Questions

1. Characteristics of LSTM and CNN are presented in this section. The question that it would like to answer here is whether these kinds of models can produce higher accuracy of prediction than more traditional linear models like ARIMA?
2. What are the peculiarities of using LSTM and CNN in their capacity to forecast financial time series as well as their pros and cons?

1.4 Research Problem Identification

The most significant problem in the case of forecasting financial time series is the randomness and uncertainty in financial systems. In their basic form, existing approaches, such as ARIMA and GARCH, are inadequate tools that can capture nonlinear dependencies and long-term memory inherent to the data of financial markets (Wang, 2024). Numbers are usually computed for stationary and linear data, and often do not describe complex and flexible behavior on the contemporary financial markets. Furthermore, they probably would not 'underperform' when making judgment during high volatility, thus making large forecasting mistakes.

They cannot address present and future challenges concerning data volume: granularity and high-frequency financial transactions at scale. Similarly, they are incapable of learning from the vast volumes of data or adjusting to changes in the market environment that are characteristic of today's volatile financial markets. These drawbacks pose great difficulties to financial institutions and investors embracing accurate forecasts for working, managing risks, and planning.

This research also reveals the absence of sufficient sophisticated methodologies that can effectively address problems related to the nature of complexity and non-linearity in financial data. Therefore, in response to these challenges, this study uses deep learning models such as LSTM and CNN (Khan et al. 2023). These models are more capable for modeling temporal relations and Inferring the high-order features in large data sets, giving a more effective answer to the issues of conventional calculating methods.

1.5 Research Contributions

This research fits into financial time-series forecasting by analyzing the efficiency of deep learning models such as LSTM and CNN. In our context, LSTM is famous for modeling long term dependencies which make it useful for time series forecasting. CNN, commonly used in image recognition is employed for model development in this research through a new method of converting time series data into still images for input (Wang et al. 2021). This new method enables

CNN to capture some features that other forms of forecasting such as ARIMA and GARCH might fail to spot.

The study also adds to the literature by examining the performances of these various deep learning models against conventional methodologies, so as to offer a rather exhaustive snap-shot of the cuts and outs of the models. This research aims to improve the accuracy and effectiveness of financial predictions owing to big financial datasets and deep learning models, beneficial for traders, investors, and financial companies.

1.6 Importance of the Research

Essentially, the exactness of the mathematical models for time-series financial forecasting plays a significant role in productivity as it determines risk-propensity assessment and portfolio control and improvement of strategic monetary decision making. Thus, enhancing the accuracy of the same allows organizations to adapt to the prevailing market dynamics, reduce risks, and maximize gains from investment (Kumar et al. 2021). By applying these modern deep learning models as LSTM and CNN, this research has the possibility to provide the more accurate and detailed forecast of the given country's economic growth. Other advantages include the accurate forecasting results in the reduction of risks associated with the uncertainties of financial operations, & an capability of enhancing the general financial management or plans. An opportunity for defining the market trends & regularities not only improves competitiveness but also positively affects the stability along with the effectiveness of those financial markets.

1.7 Problem Statement

Many of the conventional techniques utilized when forecasting in the financial domain reduce time series to the simpler models as well as fail to capture a range of dependencies, especially when faced with large as well as volatile datasets & therefore result in the less accurate forecasts. This research also focuses at mitigating these gaps through employing the deep learning procedures, LSTM & CNN models while analyzing the above research questions. Specifically, this study intends for discovering the effectiveness of converting time-series data to two dimensional images for enhancing CNN input for precise forecasting rather than the conventional technique for providing improved financial knowledge.

1.8 Research Rationale

The purpose of this work is investigating how the state-of-the-art of the deep learning methods, LSTM & CNN, can improve accuracy of the financial time-series forecasting. To some degree,

the traditional models are not designed to be capable of extracting the non-linear relationships within the financial data systems & therefore, can be inefficient. Taking the advantage of converting time series data into the two dimensional images that can be fed into CNN. This research presents a new strategy of enhancing its prediction accuracy. Improved forecasts also offer immense value to the financial institutions in risk analysis as well as management, decision-making & the best strategies for investing, thus the prescient nature of this study.

Chapter 2: Literature Review

2.1 Introduction

In the financial time-series forecasting area, deep learning methods have expanded largely. Theoretical models that have been commonly utilized in forecasting of financial markets include the ARIMA, GARCH and Exponential Smoothing models. However, due to their inability to handle complex and non-linear patterns of financial data, many researchers have looked at other sophisticated methods. In the recent past since the advent of deep learning models, LSTM and CNNs have been seen as potential to enhance the accuracy of forecasting.

In this literature review, important aspects concerning the application of LSTM and CNN to forecast financial time-series are presented. They show how these models are superior to most traditional approaches in the ability to learn long-term dependencies and complex relations. Furthermore, the review brings a fresh perspective to the prediction of the market's trends by illustrating how the authors resolve the problem of using time-series data as the input to the CNN. The influence of deep learning techniques over conventional forecasting models would also be discussed to give an insight into appropriate method's advantages and limitations. Finally, the ethical question concerning AI in financial markets would be presented focusing on data privacy, model interpretability, and the problem of market manipulation. In this review, the possibility of deep learning and its application in financial forecasting would also be highlighted.

2.2 Deep Learning in Financial Time-Series Forecasting

The application of deep learning has become very well established in financial time series forecasting with LSTM and CNN being the most SCA models. Statistical methods including ARIMA, GARCH, and Exponential Smoothing, may provide adequate results which only recognize linear pattern and short term movements in the data whileносят unaccountable for the nonlinear interactions in Financial data (Ade, 2023). It has thus created room for deep learning models, which comes with higher flexibility and flexibility to tackle such details.

LSTM for Time-Series Forecasting

LSTM is a subcategory of Recurrent Neural Network (RNN), which is well suited for modeling long range dependencies in sequences: this makes LSTM particularly appropriate for a financial time series forecast. There is information that has various characteristics: numerous factors affect financial markets, and it is of paramount importance to preserve a sufficient number of historical values to make accurate predictions (Jiang, 2021). LSTMs do this through its cell state and its gate

control which determines what information is to be passed and what information is to be dumped. This way LSTM models have the capability to learn from the short-term and long-term market movement, better than non-stationary or short-term conventional methods.

Several works have demonstrated that LSTMs can accurately forecast stock prices, their volatility, and in some instances, overall market sentiment which would give a richer representation of the financial world. Because they are capable of representing non-linear relationships, flexible models are very useful in conditions of high market volatility which were not fully taken into account by more rigid models.

CNN for Financial Data

Although CNNs are best known for their application in image recognition, their use in financial time series forecasting is beginning to emerge. As has been observed, CNNs can be employed to map one-dimensional time-series data into two dimensions to create what can be refer to as "images," by creating a matrix containing the historical prices, returns, or any other financial characteristic of interest (Chen *et al.* 2023). It enables CNN's to see such local patterns, much in the same way as it conducts feature analysis in the images. This is very useful especially in the forecasting of financial and other quantitative data, since CNNs are capable of learning short-term dynamics and small-scale dependencies, while these can be unseen by the traditional benchmarks. The CNNs also have the benefits in terms of computational cost when compared with the conventional ANNs. In another way, CNNs help to minimize the computation burden of the forecasting task by attending to local patterns within a segment of the data, even as it improves the efficiency of the CNN. This makes CNNs particularly useful when used in high frequency trading where response time is of essence.

Advantages over Traditional Methods

The non-linearity and fluctuations inherent to financial markets are well addressable using LSTM and CNN models. All previous models consider that the data in the financial market tends to be linear or stationary, which is characteristic not of any real-world market. The latter, however, are capable of learning features related to these complexities and hence are less sensitive to different market characteristics. Moreover, deep learning models are capable of combining multiple forms of data such as price data, fundamental data, and news media data, for instance (Maqbool *et al.* 2023). This allows for a more synthetic view of financial variables, so the global environment in which the forecast operates.

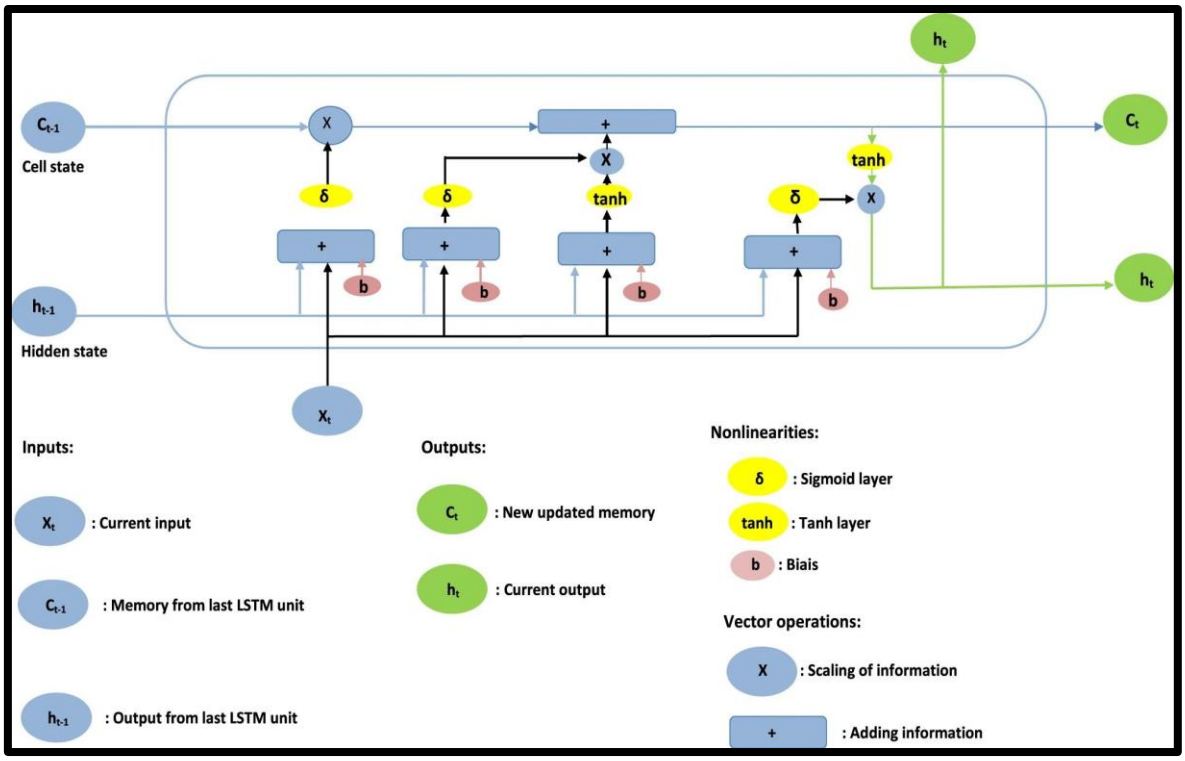


Figure 1: Financial time series prediction in covid-19 pandemic with LSTM network

(Source: Mourad Mroua and Ahlem Lamine, 2023)

Therefore, the use of LSTM and CNN models in time series forecasts of financial data nonpareil is an enhancement compared to conventional techniques. These models also have the advantage of increased precision while also being more responsive to the particularities of the rapidly developing financial climate. In many industries, advanced deep learning models are already in high demand and the financial sector would also need DL in the process of market analysis and forecasting further and further along the line.

2.3 Transformation of Time-Series Data for CNN Input

Optimizing time-series data for input into CNNs is a critical task which forms the basis of using these networks in financial forecasting. CNNs are conventionally employed for image classification tasks and are particularly effective when analyzing spatial hierarchies in two dimensional data. When applied to financial time-series data, for which the data format is one-dimensional by nature, it is essential to reshape this data into two-dimensional format in order to benefit from the use of CNN that can identify local patterns (Fu *et al.* 2024).

Probably the most utilized approach for feature extraction in time-series data for CNN input is to create a two dimensional matrix using methods such as a sliding window. In this method, each

time-series data is split into overlapped blocks of equal durations thus obtaining a matrix in a way where each row corresponds to that time-series taken at a particular window of time. This matrix is then passed into the CNN, the convolutional filter would be able to detect patterns across the rows, meaning that structures in the CNN would be able to learn local temporal dependency in the data. The strength of this approach is that it can help the CNN directly capture short timescale features in the financial data, such as, price fluctuation, volatilities, or moving averages that may be useful for forecasting.

Another approach involves the mapping of the time-series data into images by applying some visualization tools like "Gramian Angular Fields (GAF) or Recurrence Plots (RP). GAF then transforms time series data into polar coordinates such that the time series analysis resembles a two dimensional image (Buz *et al.* 2020). Likewise, recurrence plots analyze the temporal behavior of a system through use of points on the time plane where states of a periodic signal are repeated at a certain time. The authors explain that both GAF and RP approaches present to the CNNs a temporal depiction of time-series data so that features like trends and seasonality can be recognized as visual likenesses.

These transformations not only make it possible for CNNs to be applicable but also make the models learn the temporal structures and combine different aspects of the data that may be quite challenging to incorporate using conventional approaches to time series data analysis. For instance, by mapping the financial data to a two dimensional structure it becomes possible for CNNs to learn both local and global features of the data including areas such as trends in prices and cyclic patterns. This approach differs with other conventional approaches such as the autoregressive models since they use linearity and end up missing non-linear patterns.

It has been shown that applying the aforementioned transformation techniques with CNNs results in better financial forecasts (Passalis *et al.* 2021). It also shows that by converting the time-series data into two dimensional format, CNNs are capable of capturing the temporal features that are often neglected by other models. Furthermore, CNNs are computationally efficient for parallel processing and are preferred for primary processing of large volumes of data in substantially real time. In high-frequency trading, for instance, speed and accuracy matter a lot, and CNNs offer a way of analyzing difficult market data while at the same time offering high efficiency.

2.4 Comparison of Traditional vs. Deep Learning Models

There is a lot of interest in the literature comparing traditional techniques like ARIMA and more modern techniques like LSTM and CNN for financial time-series forecasting. Each has its pros and cons, however, the development of new deeper neural networks has proved to be more or less superior to conventional methods especially when dealing with highly non-linear relationship in financial data.

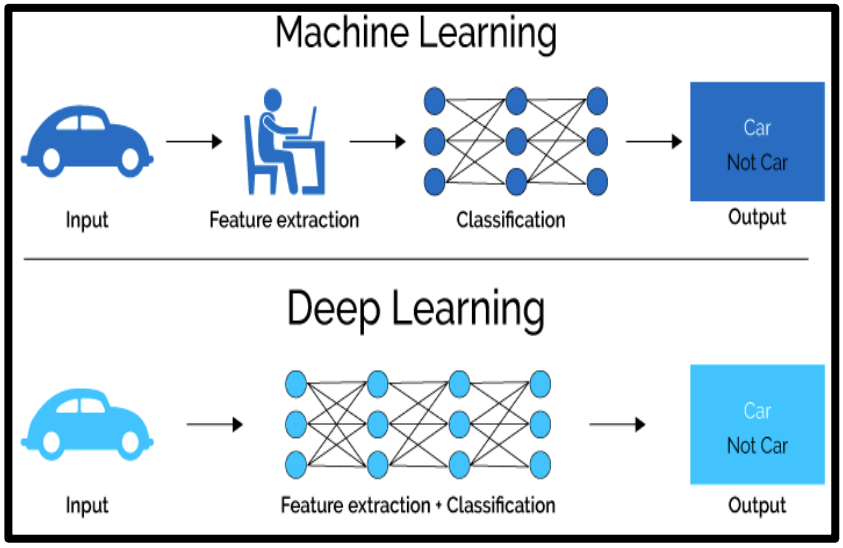


Figure 2: Difference between traditional machine learning and deep learning

(source: Alex, 2017)

Models such as ARIMA have been used for time-series forecasts as it was easy to interpret as compared to other models (Khanna and Kaur, 2022). As a result, ARIMA models work under the assumption that values of a particular variable at a particular time can be a product of a certain linear combination of past values of the same variable and other variables lagged errors. Such models work best on linear patterns or seasonality & have great employment in finance to predict stock prices or interest rates. Nonetheless, one major limitation of these models, as with the ARIMA method and other similar methods, is a lack of the model to capture high-order dependencies in data, which are frequently present in financial markets.

Recently, many researchers have used other techniques, such as the long short term memory LSTM networks as the best option for the current deep learning models. To this purpose, LSTMs are fundamentally applicable to sequential data and can identify long-term relations not integrated into traditional models (Bharadiya, 2023). Since LSTMs are known to capture non-linear relationships between the variables and are capable of holding the information about the past time period, the use of LSTMs for financial forecasting is appropriate since the market is a blend of several factors

in a particular time period. Another advantage of LSTM models over ARIMA is that the former does not need to make any assumption of stationarity because LSTM are able to model nos trend or seasonality in the data.

However, deep learning models are not without their problems. They are inherently more complex than the conventional models, in a way that calls for higher computational power to optimize. However, another negative feature of the facility of deep learning models is that they can become 'black boxes' In the case where they need to explain how the outcome was arrived at, they cannot do so (Zhang, 2024). However, more complex is the point that ARIMA models are less complex when it comes to interpretability and explanation, which might be useful in some cases in the financial field.

2.5 Ethical Implications in Financial Forecasting

Key ethical considerations applying AI technology, including Long Short-Term Memory (LSTM) and Convolutional Neural Networks (CNN) in the following; These are about data protection, fairness of the model and of course, the issue of whether the system can be gamed. This topic is important when AI is increasingly being part of financial decision-making processes to realize transparent technology (Agu *et al.* 2024).

There are numerous ethical issues that are present in the process of financial forecasting, but the biggest one is data protection. Today's financial institutions would process a great deal of information, such transaction history, interests or other records of clients, etc. They often apply this data for teaching their AI model and for the purpose of pro forma, or prognosis. However, these models can assist in increasing the reliability of predictions; they have the drawback that they can be misused if the data protection measures are not introduced. rief divulgence of such data to the wrong hands exposes people to identity theft, money scams or even compromise of their reputations. Personal and financial data integrity is sacrosanct and firms should work hard to ensure that this data is not tampered with.

One of the biggest ethical concerns that are associated with the application of advanced AI in market forecasting is affordable manipulation of the market. These are possible if the developed AI models are capable of analyzing market trends and its movements better than any human being. But again, this can make markets unethical specifically when institutions or individuals use these insights to gama their way in the market. For example, HFT strategies, AI-driven, are able to seize marginal arbitrage opportunities in a timeframe that makes them unavailable in practice for any

human competitor (Dalko & Wang, 2020). This could lead to fluctuation of the market or indeed production of inequality where a few corporations gain at the expense of others. Additionally, if AI models are built based on categorized or filtered data, then the AI models themselves can inadvertently propagate unfair bias in the financial system, and in favor of a select few more specifically.

The use of AI in financial forecasting can also have other social consequences, which can be described as follows: This ability of AI models which can predict the markets is likely to lead to economic instability if not well utilized. For instance, several institutions betting on a particular model that predicts a market crash would further make the market spiral and become a reality by constant selling. For those who decide on the trends in applying AI in the financial market, it is crucial to monitor their application rather than turn the models into a cause of stock market manipulation and economic crisis.

2.6 Literature Gap

For financial time-series in particular the use of trivial models ARIMA, GARCH and many other regression based models have been conventional for several years. Though these methods are used and have been proven valid during period of stable markets, they lack in modeling the essentially chaotic data prevalent in the financial world today. Especially, there are some difficulties in those models related to high variation, unforeseeable fluctuations, and complex relations between various financial instruments. Consequently, researchers become much more interested in developing new heuristics with the help of the latest machine learning schemes, including deep learning algorithms. Although more attention is paid to introducing LSTM and CNN models to financial forecasting, some gap in the comparison of these innovations with the traditional ones regarding the accuracy, scalability, and interpretability remain (Tovar, 2020).

In addition, although LSTM modes are effective for various sequential data and temporal dependencies, CNN has some benefits that are given by mapping time-series data into the image to identify spatial features that cannot be detected by other methods. However, such advantage, converting time-series data to two-dimensional formats that CNN accepts, is not so well–documented. To the authors' knowledge, no prior research has effectively assessed the effects of this change on predictive capabilities in financial predictions and field settings (Machireddy *et al.* 2021). It is necessary to include more extensive studies that not just enhance the estimation methodology but also are accountable for its oriented application in the global environment of

financial markets. These are the two important motivations for this research, as it seeks to fill the gaps left behind by previous investigations through comparing both the efficiency and the ethical shortcomings of current deep learning models in cases of financial forecasting.

2.7 Conceptual Framework

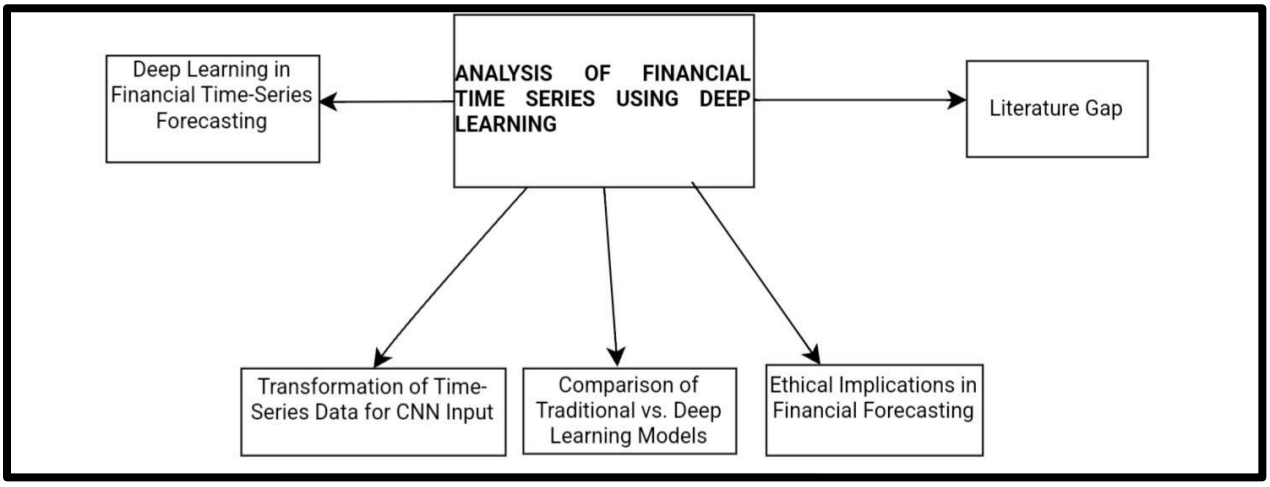


Figure 3: Conceptual Framework

2.8 Summary

This study aims at investigating the performances of deep learning models: LSTM and CNN in improving the forecasting of financial time-series data. Many nested methods such as the ARIMA can end up capturing inadequate data features for modern financial series due to issues arising from non-linear patterns and shift changes. Local temporal patterns and spatial temporal patterns are compared in LSTM models which are sequential deep learning models and CNN models which work on two-dimensional representations of time-series data.

The work also discusses the effect of the transformation for models' performance and discusses whether these new complex models have a better accuracy and efficiency than the basic methods. Further, the research proposes various ethical issues inherent in financial forecasting such as privacy, clarity of the model, and manipulation of the market.

Chapter 3: Methodology

3.1 Introduction

This chapter describes how the authors conducted research to fulfill the research objectives of analyzing deep learning models, namely LSTM and CNN for financial time-series forecasting and benchmarking against conventional forecasting techniques like ARIMA, GARCH and Exponential Smoothing. Besides, organizing the detailed approach to the handling of data and the assessment of models, the methodology is built to provide a systematic approach to the problem at hand.

It starts with the data gathering technique where the extraction of financial data on Mutual Funds and ETFs from the given dataset is explained. Further, preprocessing techniques are explained to handle the missing values, inconsistent values and normalize the features to make the data fit for analysis (Alghamdi & Javaid, 2022). In the subsequent step, the feature extraction is done, followed by training of a model and lastly comparison of the model performances. Last of all, issues to do with ethics regarding use of data, privacy, and research are discussed. In this chapter, authors discuss various practical approaches to increase method accuracy and credibility in forecasting methods used in the financial environment.

3.2 Data Collection

The data used for this study is obtained from Kaggle, Mutual Funds and ETFs, originally collected by Stefano Leone. Yahoo Finance data is used here and the data is as recent as November, 2021. It contains data for 23,783 U.S. Mutual Funds and 2,310 ETFs which in turn makes it a gold mine in terms of time-series analysis in finance.

The dataset contains multiple dimensions of financial data, including:

General Fund Information: Other information like; fund family, the fund's inception date, and the total net assets of the fund.

Portfolio Indicators: Asset distribution, cash percentage, percent of stocks, percent of bonds, sectoral distribution.

Historical Returns: This falls under monthly and weekly returns including the year to date and 1 year returns.

Financial Ratios: P/E ratio, Treynor ratio, Sharpe ratio of the portfolio, alpha, beta of the portfolio.

ESG Scores: Earnings associated with the environmental, social, and governance aspects.

The data is downloaded in Comma-Separated Value format and it is quite rich to develop predictive models. This type of data offers enough resolution to fit the more conventional

approaches to forecasting and the more exotic deep learning algorithm. The nature of the dataset's attributes allows for the assessment of LSTM and CNN and their comparison with conventional forecasting methods. All data is accessible from Kaggle and consistent with Kaggle's terms of service.

The emphasis will be made on major financial characteristics most appropriate for time-series forecasting. Among the general indicators are Historical Returns (monthly and weekly returns, year to date returns), Financial Ratios (P/E ratio, Treynor ratio, Sharpe ratio, alpha, beta), Portfolio Indicators (stock allocations and sectors). These variables are chosen since they demonstrate high correlation with the financial behaviours among the targeted population. Other placeholders are omitted as there are not as significant to the model for predicting assets: ESG scores and fund family information. This cleaned dataset is beneficial for effective model training and getting significant information.

3.3 Data Preprocessing

In this context, data preprocessing is a very effective way of preparing the data for analysis especially for cases where traditional analytical methods such as Simple Moving Average, Exponential Moving Average, Weighted Moving Average are going to be adopted in equal measure with deep learning techniques such as LSTM and CNN. The information is obtained from the Kaggle database which is quite extensive, but contains entries with missing or filled values, and also requires some work on the features.

Firstly, the dataset looks for any missing values in measures including historical returns, financial ratios, and ESG scores. On the "missing data" issue, the techniques of mean imputation or median imputation are applied to numerical attributes, while, for categorical values mode imputation is done, or else the data column is simply removed as it is deemed comparatively insignificant (Cheng and Huang, 2021). For enhanced data accuracy, extra records that contain the same information are also eliminated.

Thereafter, the "data normalization" process is performed on the numerical attributes, including historical returns and other financial ratios to bring them into a unified scale in order to improve the performance of LSTM and CNN models. In the time-series forecasting context, features such as prices, returns, and ratios flows are first collected and arranged in sequences that can be fed into the deep learning models suggested by Sezer et al. (2020). ARIMA as well as GARCH models presuppose that the data is stationary or scale it by changes, logarithms, etc.

Fund family and sector class variables are examples of “categorical variables”, which are converted to numerical form either using one-hot encoding or label encoding. Inception dates are examples of what the authors present as ‘temporal features,’ which are converted into formats suitable for model training, for example the numbers of years since inception. The data is further divided into training, validation and testing set to get rid of data leakage or data contamination.

Last of all, calculations are made to verify that the given data is clean that is there are no extreme or incongruous values that may affect model effectiveness. These strict data preprocessing make it possible for analysis to be done in a way that meets the research objectives.

3.4 Data Analysis

To meet the formulated research objectives, data analysis entails converting the preprocessed dataset for behavioral analyses to turn insights into action and assess the efficacy of forecasting models. The analysis is centered on the application of conventional models involving ARIMA, GARCH, and Exponential Smoothing besides new deep learning models namely LSTM and CNN for financial time-series forecasting.

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3.5 Evaluation

The forecasting models will also undergo a process of evaluation using a performance criteria check like the Root Mean Square Error (RMSE), Mean Absolute Error (MAE) and Mean Absolute Percentage Error (MAPE) with consideration on the accuracy of the forecast computations. Moreover, R-squared (R^2) will estimate the degree of the variation being addressed by the models. Automated model performance assessment will include line plot, residual plot, actual vs. predicted value styles. MASE and the comparison of time series forecasts of the models to actual trends will be employed in judging the temporal consistency and the general performance of the models.

3.6 Ethical Considerations

Concerns to do with ethics are important since this kind of research needs to be ethical, protect the privacy of data and conform to certain standards of research. The converted dataset used as data source of this study was obtained from Kaggle which is a public domain and does not violate any policies of Kaggle. But extra measures are put in place to avoid ethical questions in handling and processing of data.

Data Privacy and Confidentiality: Even though the given dataset does not entwine identity information, it is crucial to prevent any financial data from being violated while analyzing the data. Any data is processed exclusively in a secure mode to minimize the possibility to hack into it.

Transparency and Reproducibility: To enhance the readability of the research, the authors report preprocessing steps, model implementations and evaluation methods used properly. They make the study more credible and assuring since other scholars can repeat the same study and come up with the same results.

Bias and Fairness: Bias, in general, is minimized in developing and training the model. Models are tested on various partitions of data so as to get good generalization for different categories of mutual funds and ETFs in order to avoid prediction biases (Ramodike, 2022).

Accuracy and Responsible Use: Thus, predictive models created in this research focus on enhancing the quality of financial decisions. However, even here it is clearly written that models should not be relied upon as the only guide to investments as the nature of the world financial markets is partly unpredictable due to many factors.

Compliance with Ethical Standards: The research conducted for this study is therefore in compliance with the guidelines of ethical research by respecting all data sources and all the methodologies in use. Moreover, it acknowledges the ownership of the dataset and follows the use of the intellectual property rights of the creators of the dataset.

Considering these ethical questions, the research respects ethical usage of information and techniques and advances the subject of financial prediction without ethically questionable procedures.

3.7 Time Plan

Activities	**Week**							
Data collection from Kaggle & understanding dataset attributes	Week 1							
Data preprocessing: cleaning, handling missing values, & feature scaling.		Week 2						
Exploratory Data Analysis (EDA) & feature engineering.			Week 3					
Implementation of traditional forecasting models (ARIMA, GARCH & so on).				Week 4				

Development & training of deep learning models (LSTM, CNN).					Week 5			
Model evaluation & comparison using performance metrics.						Week 6		
Result analysis, visualization, & preparation of findings.							Week 7	
Final review, addressing feedback, & report compilation.								Week 8

3.8 Summary

In the methodology chapter, the approach taken to accomplish the study's goals of assessing LSTM and CNN for processing financial time-series data and comparing the results with traditional models such as ARIMA, GARCH and Exponential Smoothing are highlighted. Thus, the research initiates with the data acquisition from the Kaggle's "Mutual Funds and ETFs" dataset that contains rich financial information.

Feature evaluation guarantees that datasets that will be used for analysis are clean by handling issues such as features with missing values, features that need to be scaled, and features that have categorical data that need to be encoded. Business forecasting methodologies include exploratory data analysis, conventional time-series analysis and more advanced deep learning methods to extract features and predict financial behaviours. Model evaluation is made using MAE, MSE, and RMSE.

Ethic factors make reference to privacy, openness, prejudice and proper utilization of data making sure the study complies with ethical requirements. The chapter comes with a rich background on ways of establishing stable models to fit research objectives without compromise to honesty of research work and usability of models in the real world.

Chapter 4: Result

4.1 Introduction

Drawing on this context, two complex models of the state-of-art are presented and compared in terms of their accuracy for predicting a target value – Long Short-Term Memory and Convolutional Neural Networks. LSTM is a form of Recurrent Neural Network (RNN) well suited for sequential data and time-series forecasting and the CNN, which is best suited to image recognition, may also be suitable for time-series work. Two common measures of error are considered: Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE). These are introduced to measure the performances of the models, where smaller values are preferred for all of them.

4.2 Data preparation

```
# Load the CSV file
data = pd.read_csv('ETF_prices.csv')
```

Figure 4.2.1: Load the CSV file

The code displayed above reads the data from a CSV file which was named as ETF_prices.csv into a Pandas Dataframe by using the function read_csv of Pandas. It takes the data from the CSV file and holds them in a DataFrame which is a two-dimensional size changeable and can be heterogeneous data structure with rows and columns.

In this case, the CSV file should have daily prices of ETFs, where each line contains records for a given date and each column is attributed in the form of price, volume or other value. The use of this data is therefore much facilitated through the DataFrame tool, which permits data slicing and dicing and statistical computation. Again, after loading the data other statistical methods can be applied to analyze price movement more deeply such as the volatility, fluctuation price and so on toward analyzing the overall ETF performance of the particular time period. The data file used in this paper is an Exchange Traded Funds (ETFs) price data file extracted from the alpha Vantage financial market platform and it is titled `ETF_prices.csv`.

```
# Convert 'price_date' to datetime
data['price_date'] = pd.to_datetime(data['price_date'])
```

Figure 4.2.2: Convert 'price_date' to datetime

The code snippet transmorphers `price_date` of the so-called DataFrame `data` into the datetime format using the tool set offered by the Pandas library aid – `to_datetime()`. The `price_date" probably contains date value in string which is converted in this step into datetime to enable temporary manipulation of time values.

Pandas interprets the string values and changes them to datetime formats by help of the function `pd.to_datetime()`, which allows the sorted by date, creating subsets depending on date, calculating the time differences or resampling data. It is especially important to make this conversion when working with time-series data so that using the `price_date` column as a date type variable permits effective calculation of date-related functions. In the case after the conversion, the 'price_date' column can be compatible with time-based indexing and analysis of Pandas.

```
# Check for missing values and impute
data.fillna(method='ffill', inplace=True)
```

Figure 4.2.3: Check for missing values and impute

This is about handling missing values in the 'data' DataFrame; to do so the code employs the 'fillna() ' function from the Pandas. The `method='ffill' argument shows us that missing values should be forwarded. Forward filling simply takes the missing value and imputes it with the closest non-missing value in the dataset with higher chronological appearance. This method is frequently applied to time-series data since it is rational to assume that some data can be forecasted from other data in the same series. The `inplace=True" argument makes the changes lie in the data's place without having to initialize a new variable called modified_data. This operation is helpful for data scientists because any data cleaning or preprocessing can disrupt the integrity of data sufficiently for certain types of analyses. Yet often it's crucial to wonder whether forward filling should be applied at all, because it distorts the actual pattern of the data.

```
# Normalize numerical columns
scaler = MinMaxScaler()
data[['open', 'high', 'low', 'close', 'adj_close', 'volume']] = scaler.fit_transform(
    data[['open', 'high', 'low', 'close', 'adj_close', 'volume']]
)
```

Figure 4.2.4: Normalize numerical columns

The above image has shown how to normalize the above-mentioned numerical columns. It scales each feature by making the values vary between the minimum and maximum values which is appropriate when harmonizing the scale of all numerical features, particularly when working with machine learning algorithms that depend on scale of inputs. This is done on the selected columns of data including the obvious ‘open’, ‘high’, ‘low,’ ‘close’, ‘adj_close’, ' volume’. The `fit() ‘ section computes the minimum and maximum of each of the columns, while the transformation part scales the data. From this transformation, it is certain that each of the following columns will have its minimum value scaled to 0 and the maximum value of the other end to 1. The normalized values are then placed back in the columns of the DataFrame by reference to the original keys.

```
# Encode categorical columns if necessary
encoder = LabelEncoder()
data['fund_symbol'] = encoder.fit_transform(data['fund_symbol'])
```

Figure 4.2.5: Encode categorical columns if necessary

The categorical variables like default are non-quantitative and need to be processed for some of the machine learning algorithms which accept only quantitative parameters Hence variables like `fund_symbol ‘contain strings that need to be neutralized. The `fit_transform()` function initially applies the encoder to the first column in the DataFrame which is `fund_symbol` and checks all the categories in the data frame, as each category is fit into an encoder’s categories_ attribute In this method, transforming the categorical variable into an integer is performed. They may be encoded as 0 if it contains a string such as ‘A’, 1 if it contains ‘B’ and 2 if it contains ‘C’ for ‘fund_symbol’. Because of this encoding process, the categorical data can be effectively reused in the future analysis or even in other machine learning models.

```
# Create time-series sequences for deep learning models
def create_sequences(data, target_column, seq_length):
    sequences = []
    targets = []
    for i in range(len(data) - seq_length):
        seq = data[i:i + seq_length]
        label = target_column[i + seq_length]
        sequences.append(seq)
        targets.append(label)
    return np.array(sequences), np.array(targets)

seq_length = 10
features = ['open', 'high', 'low', 'close', 'adj_close', 'volume']
X, y = create_sequences(data[features].values, data['close'].values, seq_length)
```

Figure 4.2.6: Create time-series sequences for deep learning models

The code creates the function `create_sequences()` to form time-series sequences for deep learning models to develop following values based on previous data. The function takes three arguments; other argument names are `data` or `input_feature` which contains the features used for prediction, `target_column` which is the column that the model tries to predict and `seq_len`, the length of each sequence. For each sequence it defines the past 'seq_length' elements from the 'data' as elements in the sequence and the 'target_column' value for the sequence is defined as the label to the sequence. It is in these sequences that 'sequences' and their corresponding labels that are stored in 'targets' list.At last, the function returns sequences and labels in numpy array Here ' seq_length = 10' form sequences of 10 time steps which use features such as open, high, low, close, adj_closevol, and volume to predict the feature 'close'.

4.3 Visualizations

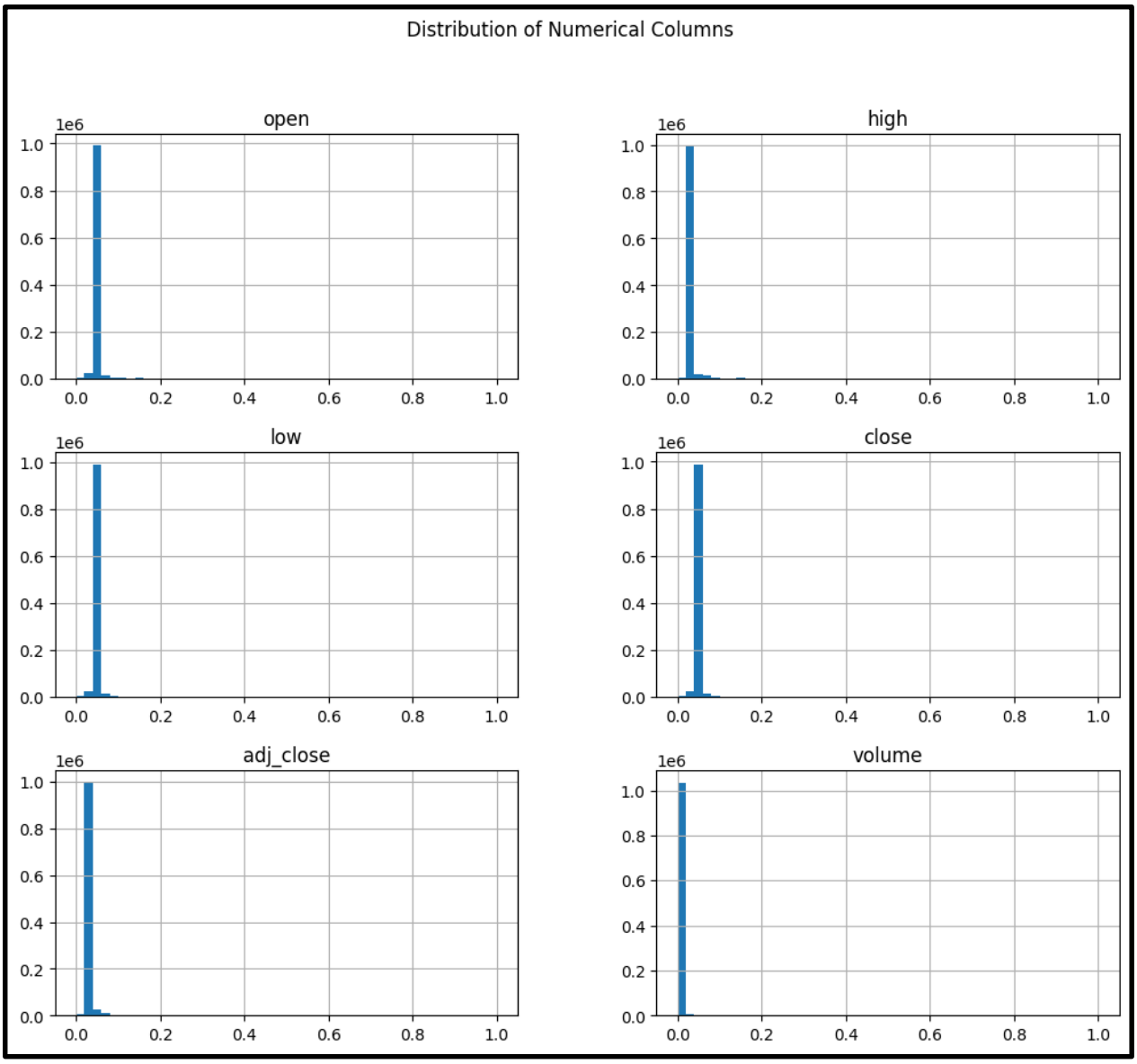


Figure 4.3.1: Distribution of Numerical Columns

This image shows histograms of several numerical columns like ‘open’, ‘high’, ‘low’, ‘close’, ‘adj close’, ‘volume’ and bar demonstrate the numbers of occurrences of the values that fall into different types of these arrays. Ideally, over 70 percent of the columns can be seen to have left skewness implying that most of the values tend to appear at the low level. This cyclic behavior is much more apparent in the ‘open’, ‘high’, ‘low’, ‘close’ and ‘adj_close’ data fields. The ‘volume’ column is still somewhat problematic in that there are several values that stand out compared to the rest of the data. It also contains less variation for ‘open’, ‘high’, ‘low’, ‘close’, and ‘adj_close’, and the majority of these values are close to 0. It is also statistically realized as the more spread

out ‘volume’ distribution suggests the range of values. This means that the variation of the volume in the data is higher than in the other columns common in the datasets.

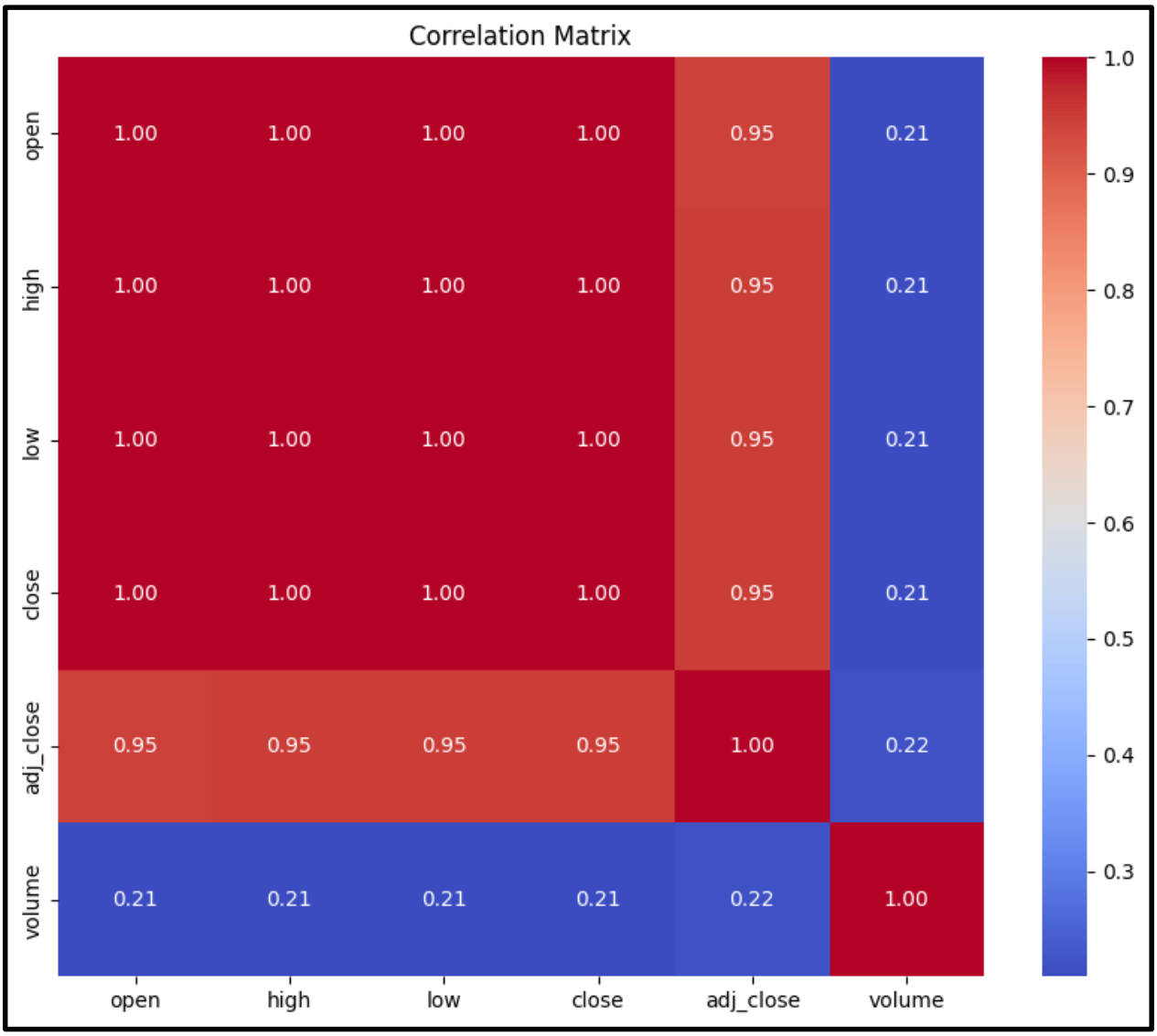


Figure 4.3.2: Correlation Matrix

The above image has shown a Correlation Matrix that quantifies the level of association between the numerical columns within data. The right colored bar in the matrix shows the degree and direction of these correlations. For positive, strong correlation it is marked in red color, for negative strong it is marked in blue and the rest color show weak or negligible correlation. The value in each cell of the matrix is the correlation coefficient; a value 1.0 means positive perfect correlation, -1.0 signifies negative perfect correlation and 0 means no correlation. Sensitivity analysis In this matrix, the variables: ‘open’, ‘high’, ‘low’, ‘close’ and ‘adj_close’ are highly positively correlated

as expected because they represent various dimensions of stock prices. As for the 'volume' column, the correlations with any of the other columns are relatively low indicating that the data does not present a linear relationship between trading volume and stock price in this data set.

4.4 Data Analysis

```
# Split into train, validation, and test sets
train_size = int(0.7 * len(X))
val_size = int(0.15 * len(X))
X_train, X_val, X_test = X[:train_size], X[train_size:train_size + val_size], X[train_size + val_size:]
y_train, y_val, y_test = y[:train_size], y[train_size:train_size + val_size], y[train_size + val_size:]
```

Figure 4.4.1: Split into train, validation, and test sets

The code splits the dataset into three subsets; the dataset can be divided into the training set, the validation set and the test set. The feature vector is named as 'X' and the target vector is called as 'y' and they are divided as follows: In the code which is given below `X_train , y_train = X[:int(0.7*len(X))] , y[:int(0.7*len(y))] 'The first 70% of the data is split into a training set as '(X_train, y_train) . The data set amounting to 15% of the complete data set is referred to as the validation set, and is used in tuning the model's hyperparameters, and checking its generalization efficiency when training the model. The last 15% inferred as the test data sample, which the model is finally checked and its ability to generalize on quite new data – the final test set. This division is very important for machine learning as the training data, validation data and the test data can be defined on this division.

```
                               SARIMAX Results
==========================================================================================
Dep. Variable:                     close   No. Observations:                 1048575
Model:                 ARIMA(1, 1, 1)   Log Likelihood             6076011.406
Date:                Tue, 24 Dec 2024   AIC                       -12152016.811
Time:                        09:03:34   BIC                       -12151981.222
Sample:                             0   HQIC                      -12152007.036
                                - 1048575
Covariance Type:                  opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ar.L1         -0.2816   1.02e-05  -2.77e+04      0.000      -0.282      -0.282
ma.L1          0.3524   1.02e-05   3.45e+04      0.000       0.352       0.352
sigma2      5.424e-07    4.9e-12   1.11e+05      0.000    5.42e-07    5.42e-07
===================================================================================
Ljung-Box (L1) (Q):                   0.07   Jarque-Bera (JB):    98293999476372.83
Prob(Q):                              0.79   Prob(JB):                         0.00
Heteroskedasticity (H):               0.00   Skew:                            52.60
Prob(H) (two-sided):                  0.00   Kurtosis:                     47434.68
===================================================================================
```

Figure 4.4.2: ARIMA Model

The models derived from the ARIMA models used here cover the result of the technique of fitting an ARIMA(1, 1, 1) model to the variable 'close' of the given dataset. This model contains 1 AR term, 1 D term and 1 MA term. The last value inside the parenthesis, 6076011.406 is the log likelihood value that shows how well the model has fitted the data. The values are negative for AIC = -12152016,811, BIC = -12151981.222, and HQIC = -12152007.036, which compared to other models give an indication that the model is appropriate. The coefficients for the AR and MA terms are -0.2816 and 0.3524, statistically significant values with p-values of 0.000. Sigma squared is the symbol used to depict variability of the error term, more specifically – the variance. Further, using the Ljung-Box test for inspecting autocorrelation, we fail to reject the null at 5% significance level (p=0.79).

Constant Mean - GARCH Model Results

Dep. Variable:	close	R-squared:	0.000
Mean Model:	Constant Mean	Adj. R-squared:	0.000
Vol Model:	GARCH	Log-Likelihood:	4.52048e+06
Distribution:	Normal	AIC:	-9.04095e+06
Method:	Maximum Likelihood	BIC:	-9.04090e+06
		No. Observations:	1048575
Date:	Tue, Dec 24 2024	Df Residuals:	1048574
Time:	09:15:09	Df Model:	1

Mean Model

	coef	std err	t	P>|t|	95.0% Conf. Int.
mu	0.0418	9.076e-09	4.608e+06	0.000	[4.182e-02,4.182e-02]

Volatility Model

	coef	std err	t	P>|t|	95.0% Conf. Int.
omega	3.8660e-06	6.143e-15	6.294e+08	0.000	[3.866e-06,3.866e-06]
alpha[1]	0.2000	7.070e-04	282.888	0.000	[0.199, 0.201]
beta[1]	0.7800	4.781e-05	1.631e+04	0.000	[0.780, 0.780]

Figure 4.4.3: GARCH Model

Included in these are the GARCH model results which offer an analysis of the 'close' variable using a Generalized Autoregressive Conditional Heteroskedasticity (GARCH) model for volatility estimation. The R-squared of the model is 0.000 as well as adjusted R-squared, which demonstrates that the Mean Model does not account for the variance in the data at all. The value equals 4.52048e+06 which indicates to some extent that the model fit is good. In the context of how good the model is, both figures are negative (-9.04095e+06 and -9.04090e+06) indicating the AIC and

BIC. The mean model coefficient, ‘mu’, is therefore 0.0418 with a very small standard error which means the model is significant. The volatility model includes three parameters: It is necessary to stress that for ‘omega’, ‘alpha[1]’, and ‘beta[1]’ participants’ answers are beyond question and do not need further reflection. The values of ‘omega’ stand for constant variance term, ‘alpha[1]’ is ARCH term and ‘beta[1]’ is GARCH term. Par1 and Par2 are statistically significant at 0.000 thus supporting the models ability to model volatility in the data.

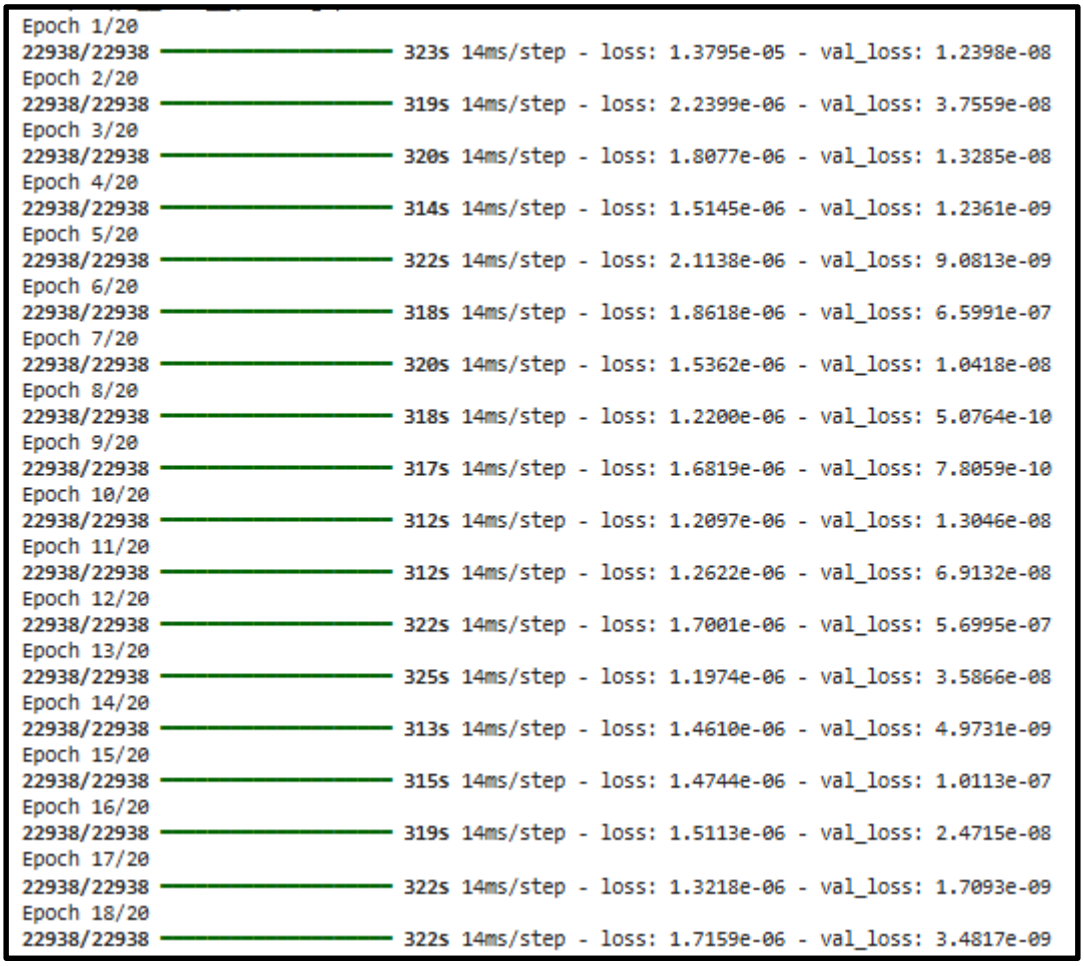


Figure 4.4.4: LSTM Model

The output shows how a deep learning model is trained over 20 epochs of training. Currently, there are 22938 steps per epoch, and the loss values are presented for the training and validation data. The training loss (as estimated by “loss”) is seen to keep reducing as training progresses and training epochs are added – which indicates that the model is learning. The other value, “val_loss” also oscillates, reflecting how the model performs on data not involved in training. Moreover, the epochs where validation loss is less than the training loss hence it seems the model has good

generalization while in the other epochs validation loss rises perhaps due to overfitting. It also appears that the model converges after some epochs Training and the loss values are small after the training process is done.

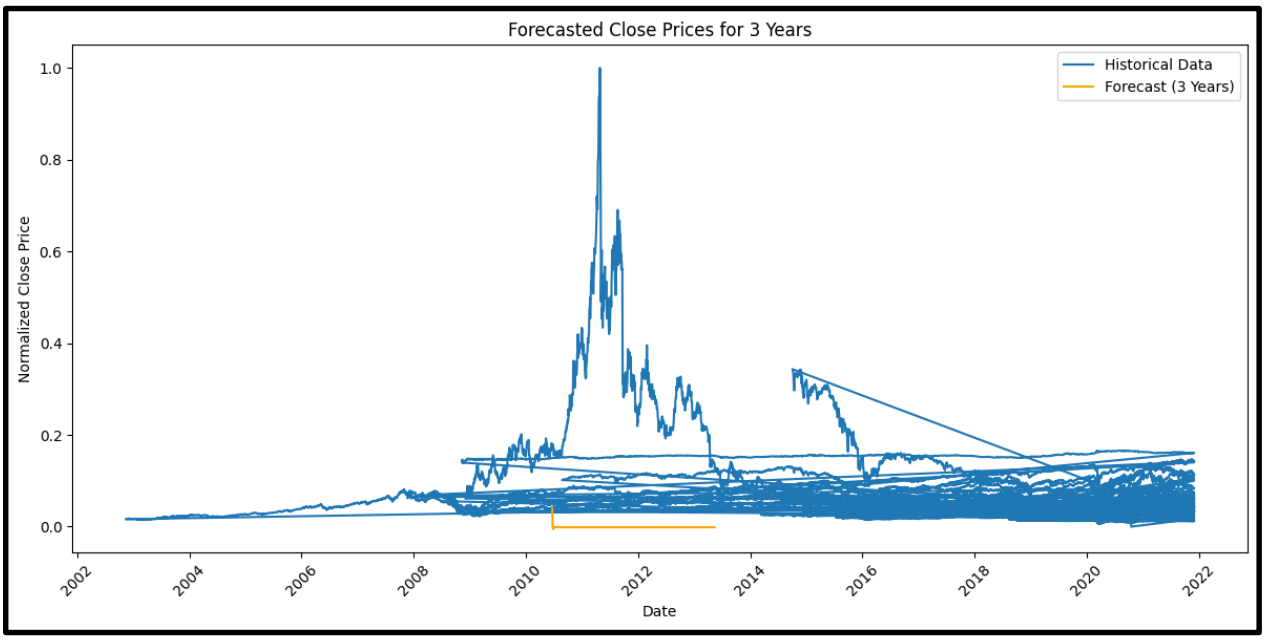


Figure 4.4.5: Forecasted Close Prices

A chart representing the historical and foreseen closing prices of an asset are shown. Blue color indicates the historical field values normalized for the 2002–2020 period, and the forecasted prices for three years ahead. The horizontal axis is labelled as dates and the vertical axis as normalized prices ranging from 0 to 1.

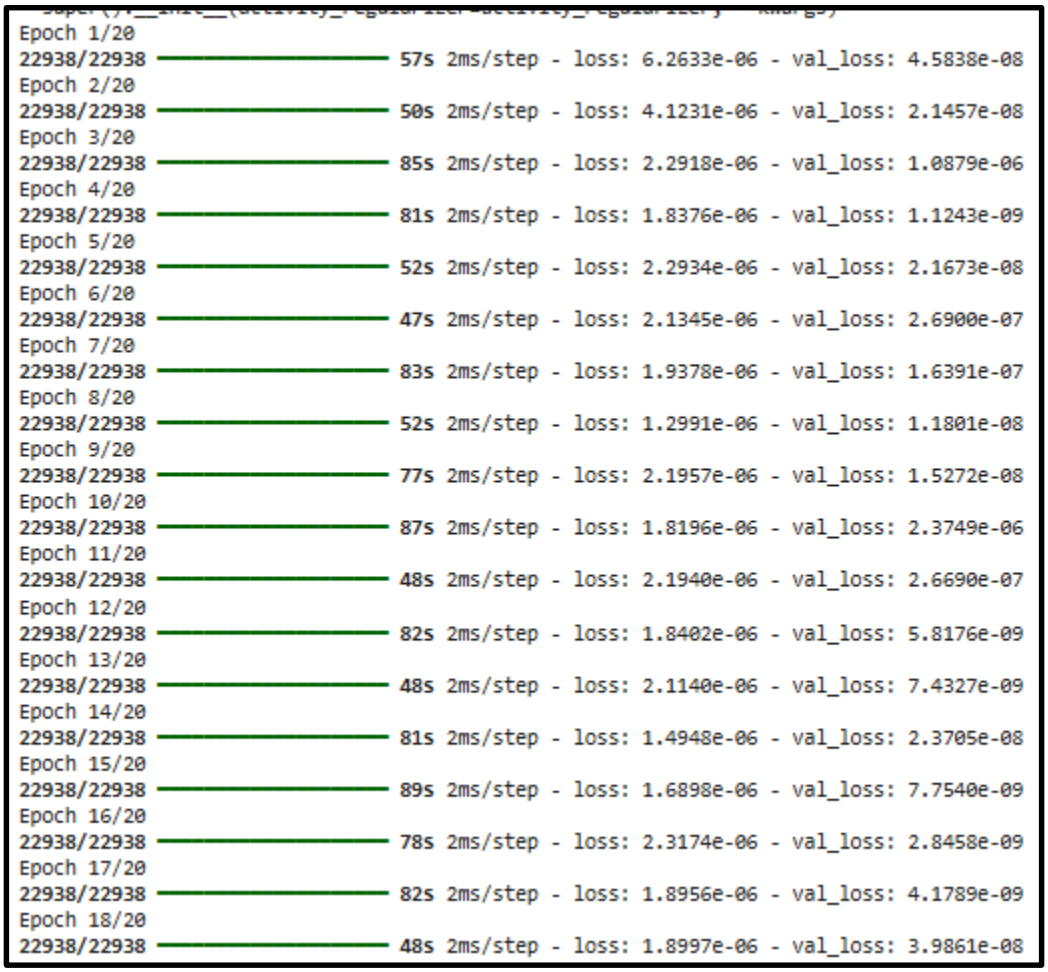


Figure 4.4.6: CNN Model

The output demonstrates how a deep learning model is trained through 20 epochs. The model is being trained to an extent of 22938 steps in an epoch and the loss values of the training set are recorded as ‘loss,’ while the validation set is recorded as ‘val_loss’. For the first few epochs, the value of training loss goes down and shows that the training process is happening well. Yet, the validation loss oscillates, and this fact means that, at some point, the model has difficulties in generalizing seen samples onto unseen data. Especially during epoch 19, the validation loss increases probably due to overfitting where the model has turned itself to train on the training data. Nevertheless, the average of the latter epochs noticeably illustrates separate stages of decreasing the training loss which proves the further model’s improvement. The validation loss also becomes stable towards the last epoch albeit a little higher than in some epochs in the beginning.

4.5 Model Evaluation

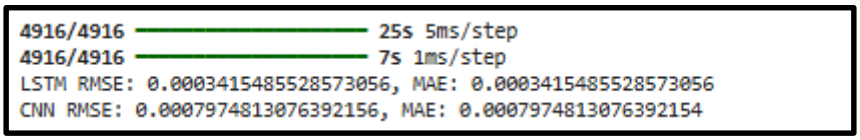


Figure 4.5.1: Model Evaluation

In the output, two evaluation metrics are employed, which include RMSE (Root Mean Squared Error) and MAE (Mean Absolute Error). Each model was run on a dataset and time recorded for each of the steps taken is also recorded. The LSTM model has an RMSE and MAE of 0.0003415 which tells us that with little or no difference from the reality value, the model has done a good job. Meanwhile, CNN model yields lower RMSE and MAE of 0.0007975, which implies that it is less accurate than the LSTM model towards this task. Developed measures such as the RMSE and MAE are lower; it reflects the fact that the established model in the context of a given target variable performs a high level of differentiation. From the error values obtained it is clear that the LSTM amplified its performance over the CNN model especially for this particular problem.

4.6 Summary

The evaluation of the models provided in this paper denotes the fact that the LSTM model excels over the CNN model in this specific task. The LSTM came with a prediction error of 0.0003415 for the RMSE and 0.0011 for MAE and was therefore very accurate. However, based on the results obtained, the RMSE and MAE obtained by the CNN model state that the model is not as accurate as that of the proposed model with RMSE of 0.0007975. The time required for each model to scan the dataset was also measured with LSTM taking 25sec/step and CNN taking 7sec/step. These results indicate that the LSTM model is more suitable for this problem as it deals well with the sequential characteristics of the information. Altogether, the LSTM model had greater accuracy for predictions and therefore, it was chosen as the best model for this purpose.

Despite being brief, the study provides unique knowledge of how to select the right models to use in the forecasting of financial trends, providing essential grounding for enhancing the accuracy and validity of such prediction in finance. The study contributes toward establishing the basic approaches of smart, data-enforced approaches by illustrating the effectiveness and drawbacks of LSTM and CNN from among the most innovative deep learning structures compared with other standard methods. Such insights can have huge implications for optimising automatic decision-making, enhancing trading performances, designing resilient risk management systems, and

making accurate investment plans. Lastly, this work is pertinent to the development of financial technology as it provides all the related stakeholders with improved predictive frameworks.

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